Using Artificial Intelligence for Campus Food Delivery

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Table of Contents

What is AI & Robots – pg. 3-4

Business Application – pg. 5-6

Competitive Advantage – pg. 7

AI & Neural Network Implementation – pg. 8-9

Operating Model – pg. 10

4WD Smart Car Information pg. 11-14

- Motor Control – pg. 11
- Light Tracing Car – pg. 12
- Ultrasonic Obstacle Avoidance – pg. 13
- Infrared Line Tracking – pg. 14

Demonstration – pg. 15

Legal Considerations – pg. 16

Ethical Considerations – pg. 17

Learning Resources – pg. 18

Conclusion – pg.19

References – pg. 20

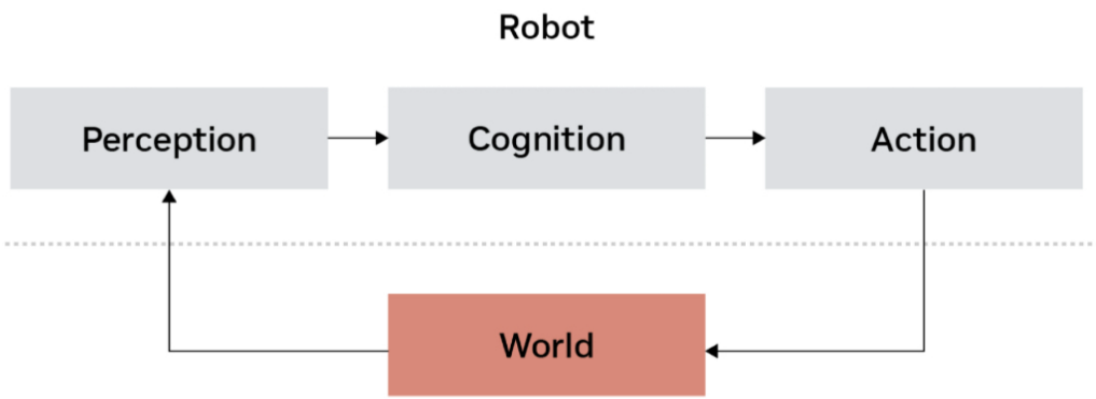
What is AI? What Are Robots?

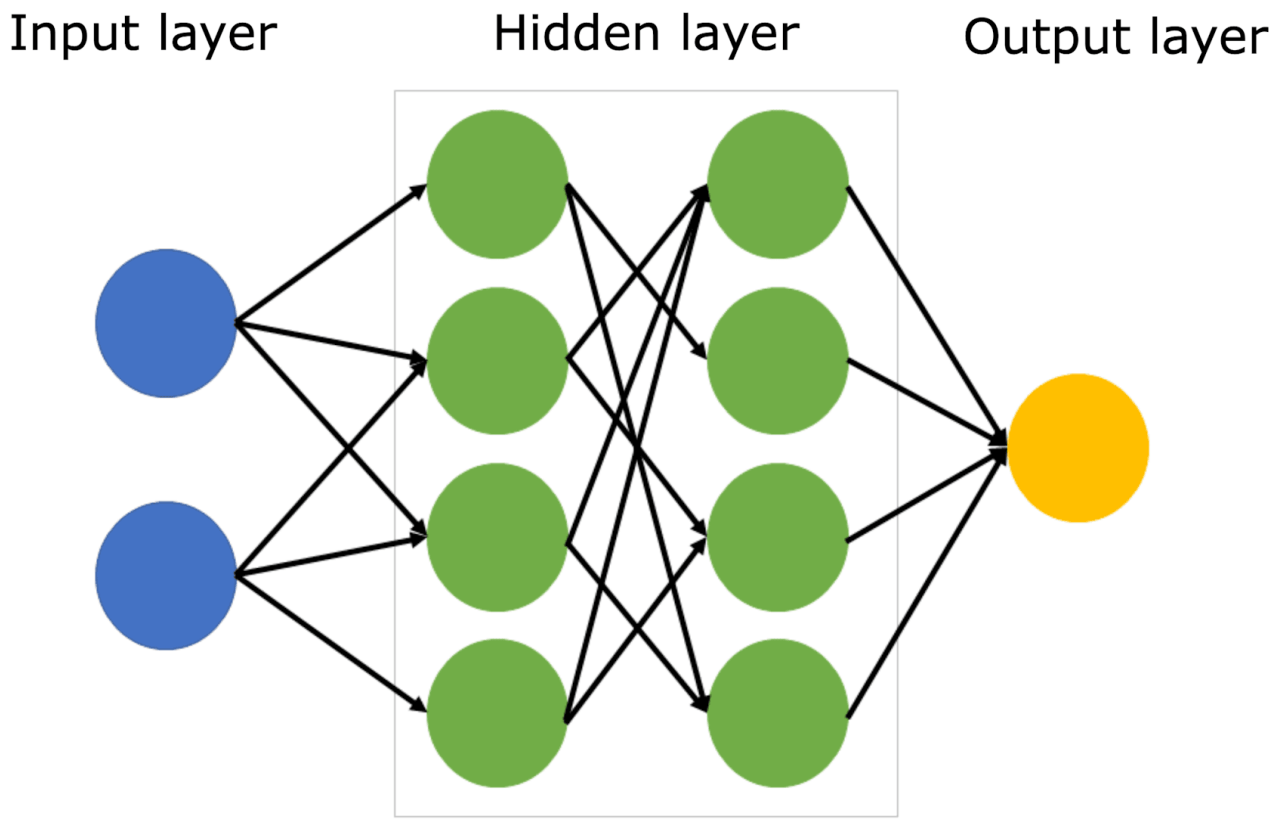
Robots:

AI agents that utilize hardware and software to sense the environment, process that information, make decisions, and act on those decisions.

AI & Artificial Neural Networks:

Computer systems designed to mimic how the human brain processes information. ANNs use artificial neurons to analyze data, identify patterns and make predictions.





Artificial neural networks

More on Artificial Neural Networks

A neural network learns to map inputs to outputs by adjusting weighted connections between simple units (“neurons”). During training, it compares its predictions to the correct answers, measures the error, and tweaks the weights (via backpropagation) to reduce that error over many examples.

Business Application

Autonomous, On-Demand Food Delivery

AI-powered vehicles autonomously navigate college, medical, and corporate campuses to deliver meals, snacks, and beverages directly to students, employees, and medical staff. Users place an order, and the vehicle delivers.

Value Creation:

1. Convenience for End Users

- Saves time for students, employees, and medical staff
- Reduces long lines and wait times
- Offers reliable access to food even during late hours or busy periods

Business Application Cont.

2. Operational Efficiency for Vendors

- Increased order throughput
- Reduced delivery labor costs
- Streamlined pickup and delivery workflows

3. Cost Reduction

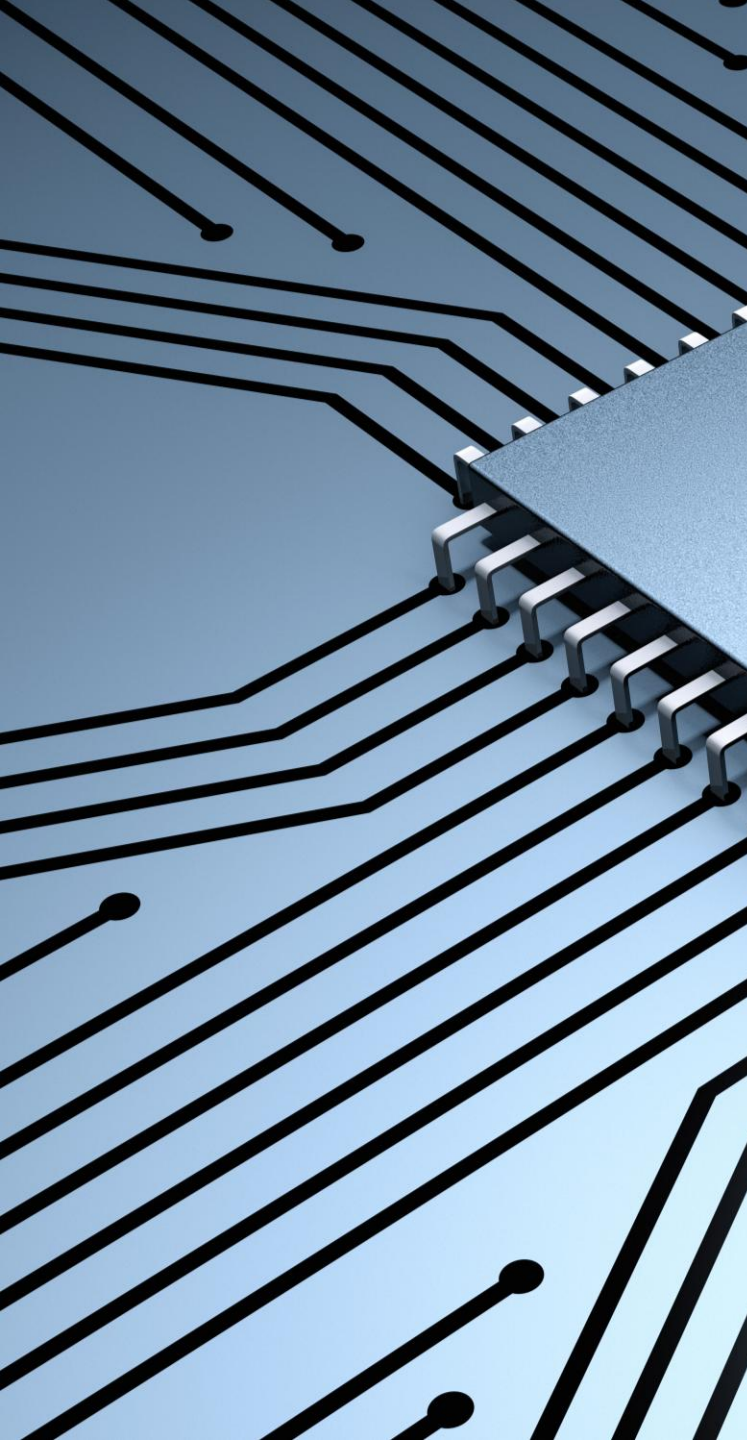
- Removes the need for human couriers
- Lower operating expenses
- Predictable cost structure
- No labor scaling needed

4. Safety & Reliability

- Reduced risk of accidents vs. bikes or scooters
- Consistent delivery performance
- Better service during bad weather or late-night hours

Competitive Advantage

Competitive Advantage	Description
Compact Size	Small vehicles navigate sidewalks, courtyards, and narrow pathways easily, reducing congestion and human traffic.
Fully Autonomous AI Navigation	Uses sensors, computer vision, and real-time mapping to drive, avoid obstacles, and deliver without human control.
Low Operating Costs	Electric, lightweight, and autonomous operation reduces labor, energy, and maintenance costs compared to human couriers.
Safe & Contactless Delivery	Controlled speeds, smart obstacle avoidance, and secure compartments ensure safe movement and hygienic delivery, ideal for medical and crowded campuses.
Scalability	Easy to integrate into existing systems such as food or cafeteria apps. No need for human labor scaling.



AI and Neural Network Implementation

Perception (CNN)

Using image recognition for:

Object Detection & Tracking

Semantic Segmentation (Distinguishes sidewalks, grass, roads, etc.)

Localization & Mapping (RNN)

Combine sensor data to build maps

GPS integration

AI and Neural Network Implementation Cont.

Path Planning & Decision-Making

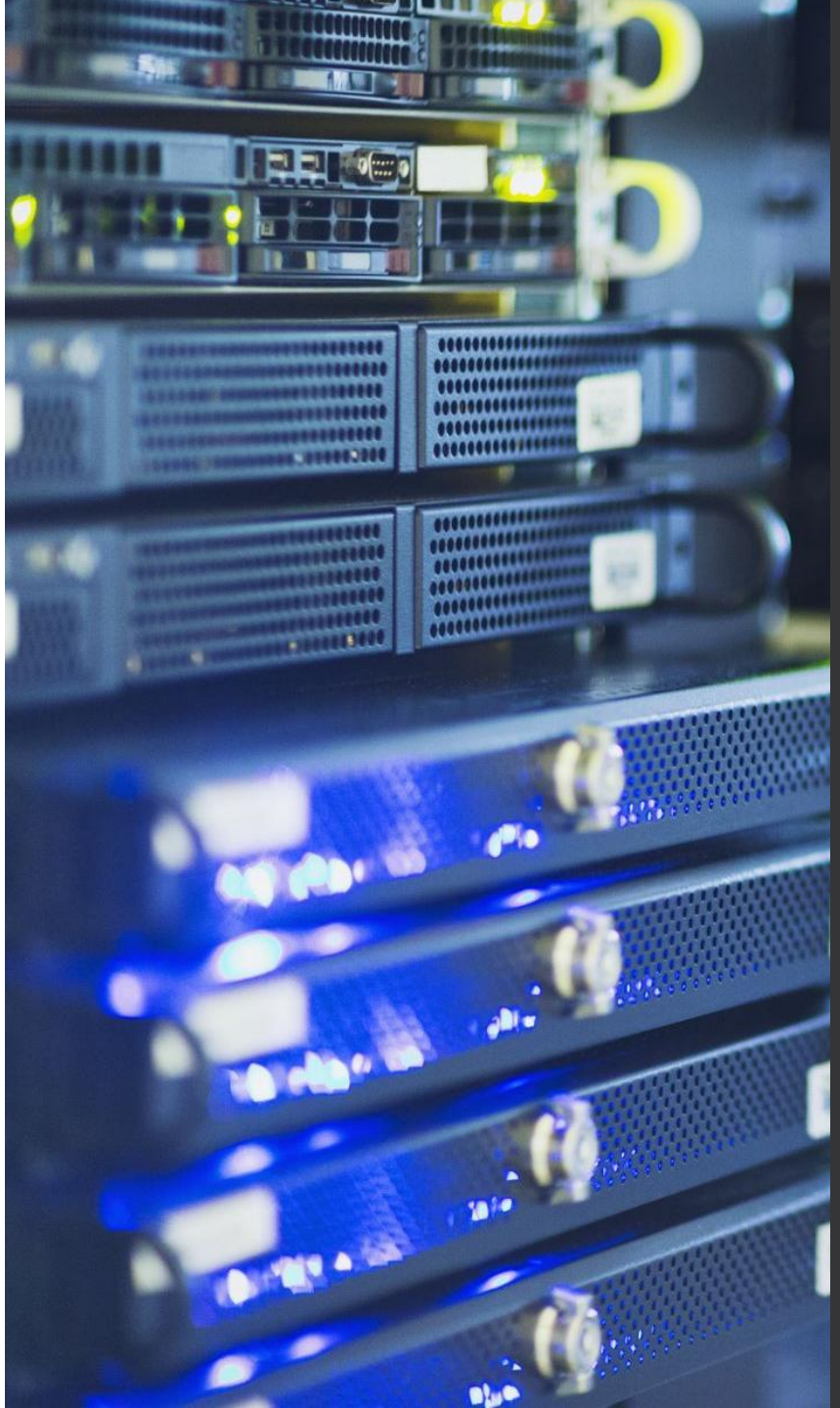
AI is used to determine the best route and how to respond to dynamic conditions.

Reinforcement Learning

- Trains the system to choose actions that maximize safety and efficiency.

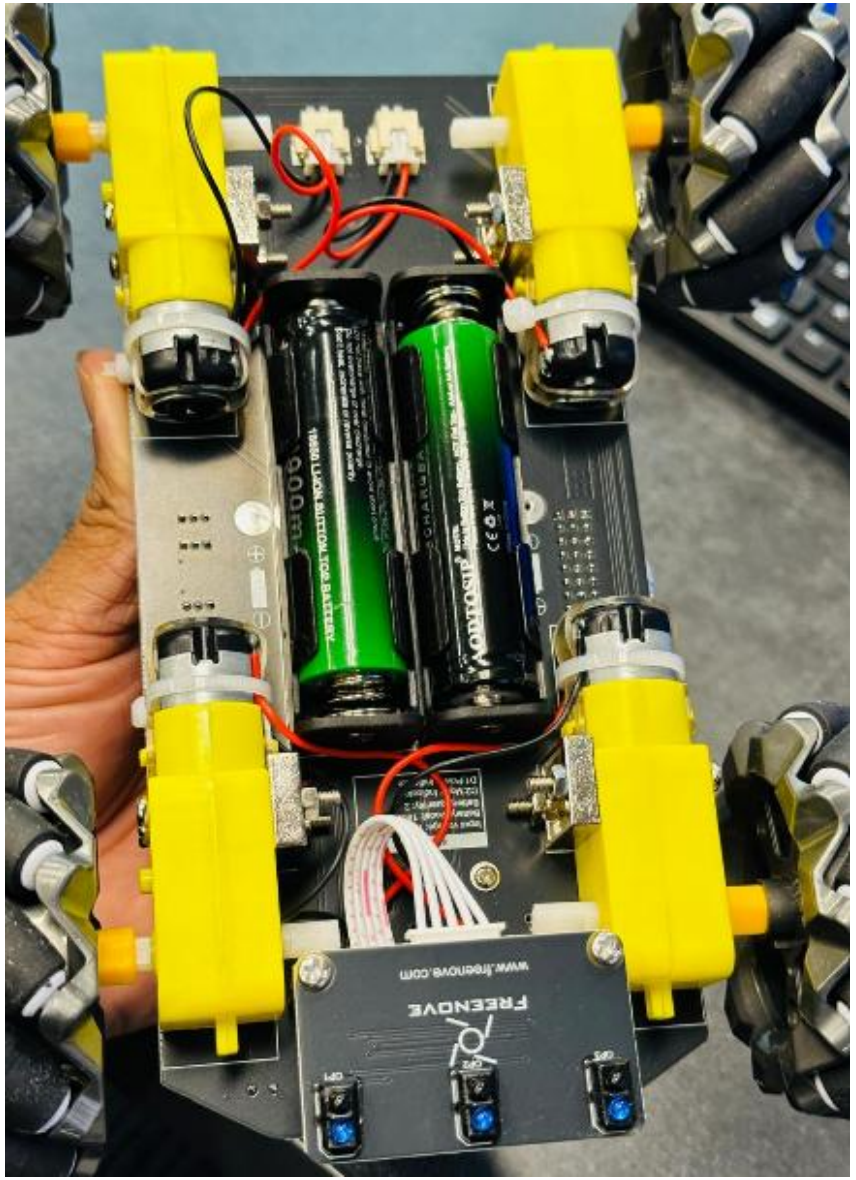
Behavior prediction

- Predicts future movement of pedestrians or cyclists and adjusts trajectory.



Operating Model

Inputs	Process	Output	Value Creation
• Visual information • Geographic data • Sensor data	• Semantic segmentation • Object detection • Mapping • Reinforcement learning • Behavior prediction	• Decision making • Path planning • Map building • Behavior based actions	• User convenience • Vendor efficiency • Cost reduction • Reliability



Motor Control for Turning the Car to One Side

How the Motor System Works

- The function `setMotorModel(A, B, C, D)` controls four motors:
  - **A** – Left Front
  - **B** – Right Front
  - **C** – Left Rear
  - **D** – Right Rear
- **Positive values (0 to 4096):** wheel rotates **forward**
- **Negative values (-4096 to 0):** wheel rotates **reverse**
- **Larger value = faster speed**; **0 = stop**

Light Tracing Car

The car always moves toward the brighter light.

How It Works

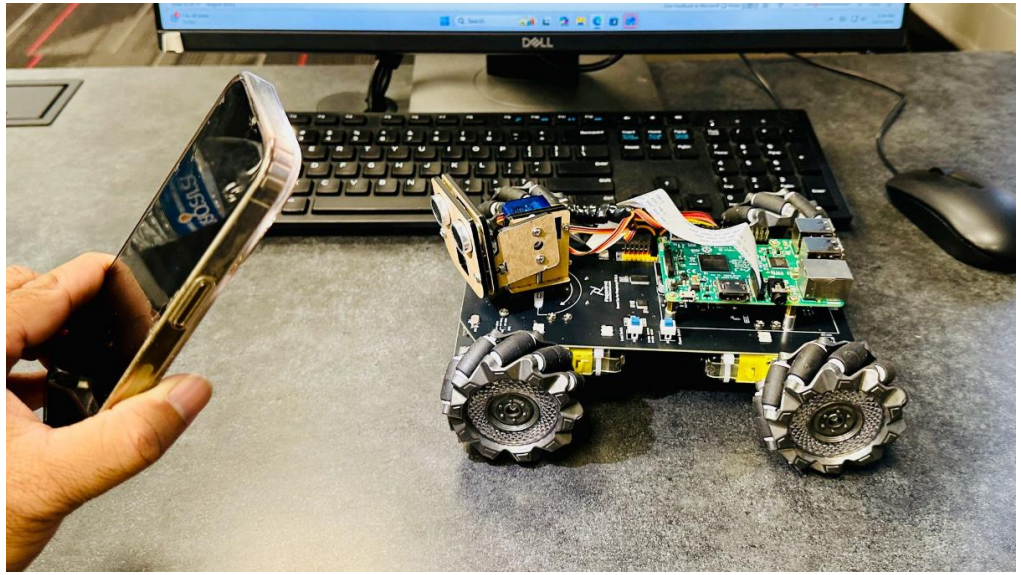
- The car has **two light sensors** in the front:

  left and **right**.

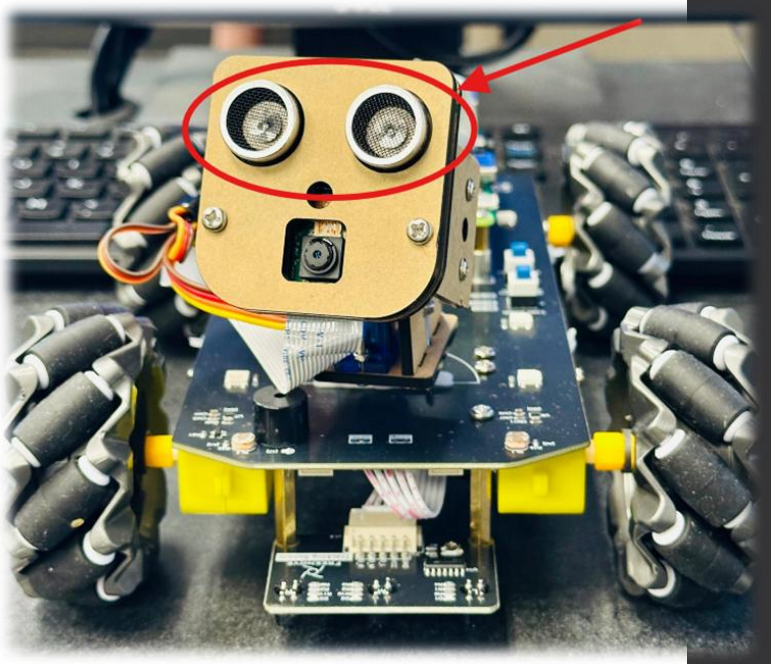
- When **more light hits a sensor**, its **voltage goes up**.
- The car compares the left and right voltages to see where the light is.

Car Movements

- Both sensors < 2.99V → Go Forward
- Left sensor > Right sensor → Turn Left
- Right sensor > Left sensor → Turn Right



Ultrasonic Obstacle Avoidance Car



How It Works

The car uses an **HC-SR04 ultrasonic sensor** mounted on a **servo motor**.

The servo rotates the sensor to **three angles**:

- **30° (Left)**
- **90° (Middle)**
- **150° (Right)**

At each angle, the sensor measures **distance to an obstacle > 30**.

Car Movements Based on Distance

Distances are checked at **Left, Middle, Right** positions:

- All > 30 cm → Move Forward (No obstacle nearby)
- All < 30 cm → Reverse + Turn Around (Dead end)
- Left < 30 cm, Middle < 30 cm, Right > 30 cm → Turn Right (Obstacle on left side)
- Left > 30 cm, Middle < 30 cm, Right < 30 cm → Turn Left (Obstacle on right side)

Simple Idea

The ultrasonic sensor scans left, middle, and right.

The car scans the area and **avoids anything closer than 30 cm**

Infrared Line Tracking Car

The car moves based on which sensors see the line.

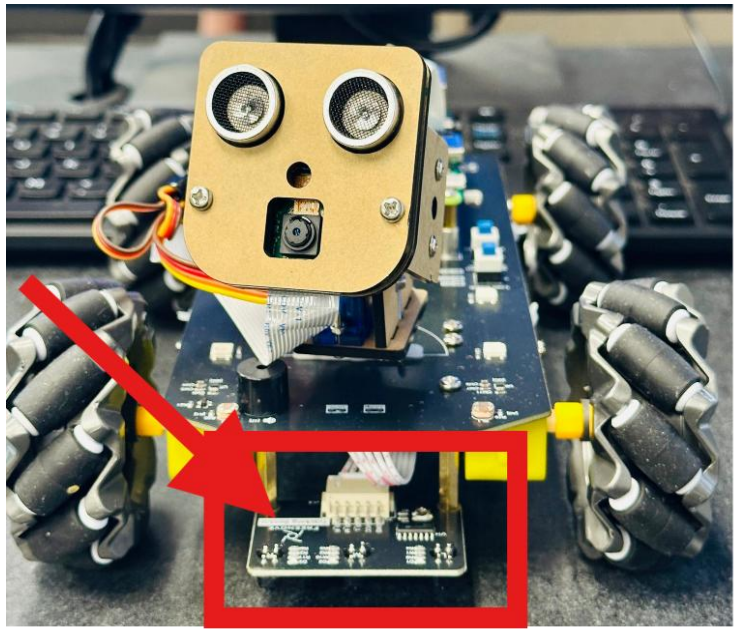
The car sensors track the black line and follow the direction

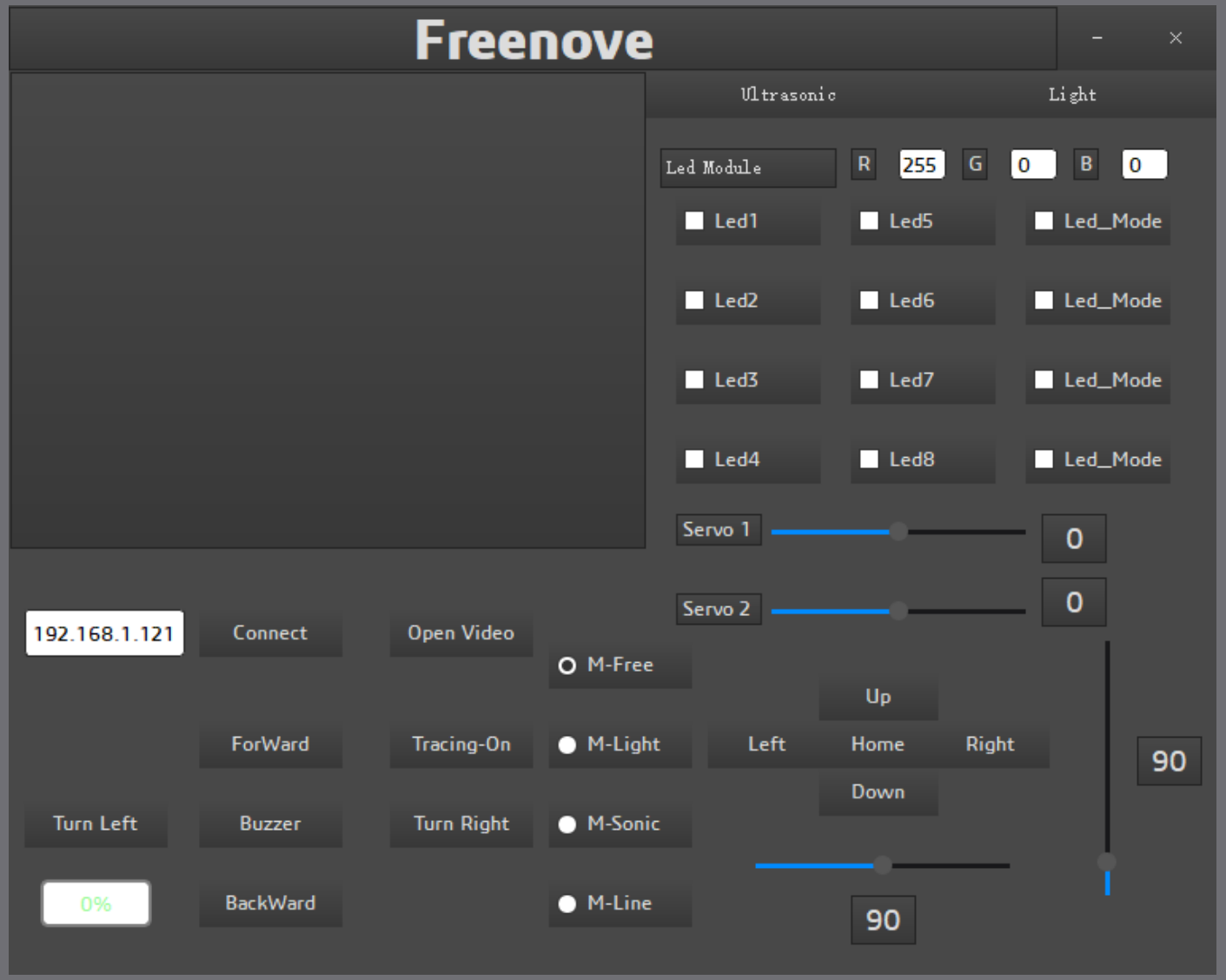
How It Works

- The car uses **3 infrared sensors** (Left, Middle, Right).
- Sensors detect the **black line** on the ground:
  - **Black = High signal**
  - **White = Low signal**

Car Movements (Rules)

- Left = High → Turn Left Slightly
- Left + Middle = High → Turn Left
- Middle = High → Go Straight
- Right = High → Turn Right Slightly
- Middle + Right = High → Turn Right





Operational Demonstration

FreeNove 4WD Smart Car movement and operation

Legal Considerations

1. Liability & Accident Responsibility

Clarify who is legally responsible if the robots:

- Collides with a pedestrian
- Damages property

2. Food Safety & Handling Regulations

Robots must comply with:

- Temperature control rules
- Contamination prevention rules
- Health department requirements

3. Safety & Accessibility Compliance

Ensure the robots do not block:

- ADA-compliant ramps
- Entrances & Exits

Ethical Considerations

1. Safety of Pedestrians

Autonomous behavior must prioritize human safety, especially in:

- Crowded student areas.
- Hospital corridors.
- High-traffic corporate campuses.

2. Potential Job Displacement

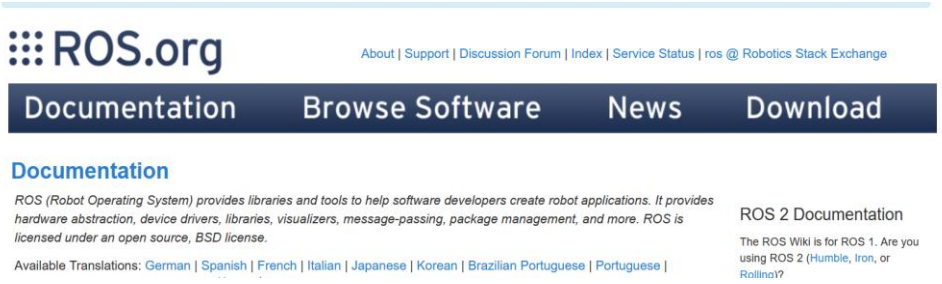
Introducing autonomous delivery may reduce the need for delivery workers.

3. Surveillance Concerns

Sensors and cameras may unintentionally capture:

- Faces
- Activities
- Sensitive locations

If you want to learn robotics where to start ?



docker
for beginners

Conclusion

THANK YOU

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